Planetary Weather Phenomena

A look into weather on planets in our Solar System and a comparison with terrestrial weather

Samuel Wigley
Student ID: 21009448

Keele University

PHY-30017

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Abstract

This dissertation takes a look through three different planets in our Solar System, at a range of distances from Earth, and the weather that occurs on them, these being the dust devils on Mars, lightning on Jupiter, and the formation of the Great Dark Spot on Neptune. An outline of the planet's atmospheric composition is discussed, along with the physics behind the generation of the corresponding weather phenomena, and ending each section with a discussion of the current research into the phenomena, and how the evolution and understanding of the planetary phenomena reflects the phenomena we experience on Earth.

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1 Introduction

Our Solar System has a variety of stellar objects that reside in it, be it stars like our sun, satellites like the Moon that orbits the Earth or one of Saturn's many moons, or planets like the one we live on. Each of these different objects is affected by weather phenomena.

Weather, and in turn weather phenomena, is a term that refers to a change or development of short-term atmospheric conditions within a given location or area. These changes are driven by a multitude of factors, like the interaction of charged particles, developments of solar energy build-up on the atmosphere, and interactions of atmospheric gases or the very properties of the area's surface and structure.

Atmospheres differ from planet to planet in the solar system but generally follow the same structure: starting at the core of the planet, the troposphere, the stratosphere, the mesosphere, the thermosphere and the exosphere. Most of the bodies discussed through this work will only involve the first two sections. The inner planets (often referred to as the "terrestrial" planets) like Venus have a thick, dense, and chemically simplistic atmosphere of nearly pure CO_2 (Taylor, 2010) and Venus specifically has an optically dense atmosphere due to having a very high albedo. Mars will be discussed in depth in a later section.

Compared to Jupiter and Neptune, the outer planets have more chemically diverse atmospheres, with Jupiter having a primary mix of H_2 and He, and a lower concentration of NH_3 and CH_4 around $\leq 1\%$ (Taylor, 2010), and Neptune, similarly to Mars, will be discussed in a later section. In stars, like the Sun, the composition is different to planets, being a gas giant with an approximate chemical composition of 73% H_2, 25% He and the remaining 2% is a mix of compounds formed during nuclear fusion (OpenStax et al., 2017). Both Jupiter and Neptune will also be covered in more detail later.

While a lot of information and studies have been conducted on the weather phenomena here on Earth, many astrophysicists have taken the leap into studying these phenomena on planets and stellar bodies through our solar system - finding that while the causes may vary body-to-body, the underlying process will remain the same. Understanding the effects and development is important when looking at these bodies, as it helps to confirm any understanding of the weather we have on Earth, and, in some cases, helps astrophysicists comprehend the structure, formation, and as of recent interest, the habitability of the body.

In particular, this dissertation sets out to answer the following questions.
What weather phenomena happen within the Solar System? and *What research has been done into these phenomena?*

2 Neptune: Dark Spot

2.1 Neptune - Planetary Overview

Neptune, the planet in the farthest part of the solar system, with a mass of 17 $M_\oplus$ (Helled et al., 2020) is approximately four times the size of Earth (Helled et al., 2020), and lies a distance of 30 AU away from the Sun (Helled et al., 2020). Originally, Neptune's existence was inferred by two mathematicians, Urbain Jean Joseph Le Verrier and John Couch Adams - from France and England respectively - however, the planet was officially discovered by Johann Galle and Heinrich D'Arrest in 1846, at the Berlin Observatory (Moore, 2014).

The planet is known as an "Ice Giant" by institutions such as NASA, despite the main core of Neptune being "rocky" in nature and its atmosphere having a largely gaseous structure, mainly made of H_2, He and CH_4 in the stratosphere, and NH_3 in the troposphere (which is what is presumed to cause the bright clouds seen in images) (Taylor, 2010). This composition can be seen in Figure 1.

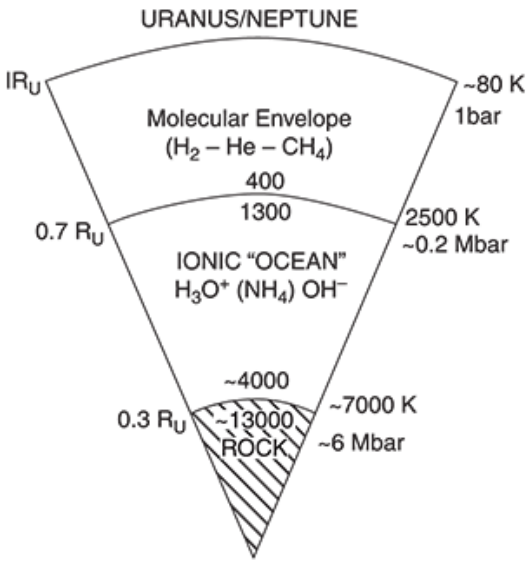

Figure 1: A schematic diagram of a sector of Uranus/Neptune that represents the chemical composition in each area of the atmosphere. Taken from Lissauer & Pater (2019).

Neptune's methane clouds have been found to be moved around the atmosphere as a result of the planets incredibly high winds, which in some cases reach up to 1.5 times the speed of sound, or approximately 1100 mph (Sharghi, 2013). Through missions like

the Voyager 2 mission from NASA, these winds have been identified as potential parts of storms in Neptune's atmosphere - an example of one of these such storms is the storm labelled "the Dark Spot".

2.2 The formation and evolution of Neptune's Dark Spot

On August 20th, 1977, NASA launched the Voyager 2 mission (Aeronautics & Administration, 2024), which set out as a flyby mission to the outer reaches of the solar system, to gather information on the outer planets; Uranus and Neptune.[1] Voyager 2 reached Neptune on August 25th 1989 after a 4.3 billion mile journey (Aeronautics & Administration, 2024), and took images that revealed a storm cloud in the atmosphere, dubbed the "Dark Spot". A more recent image of a dark spot on Neptune from the Hubble telescope can be seen in Figure 2.

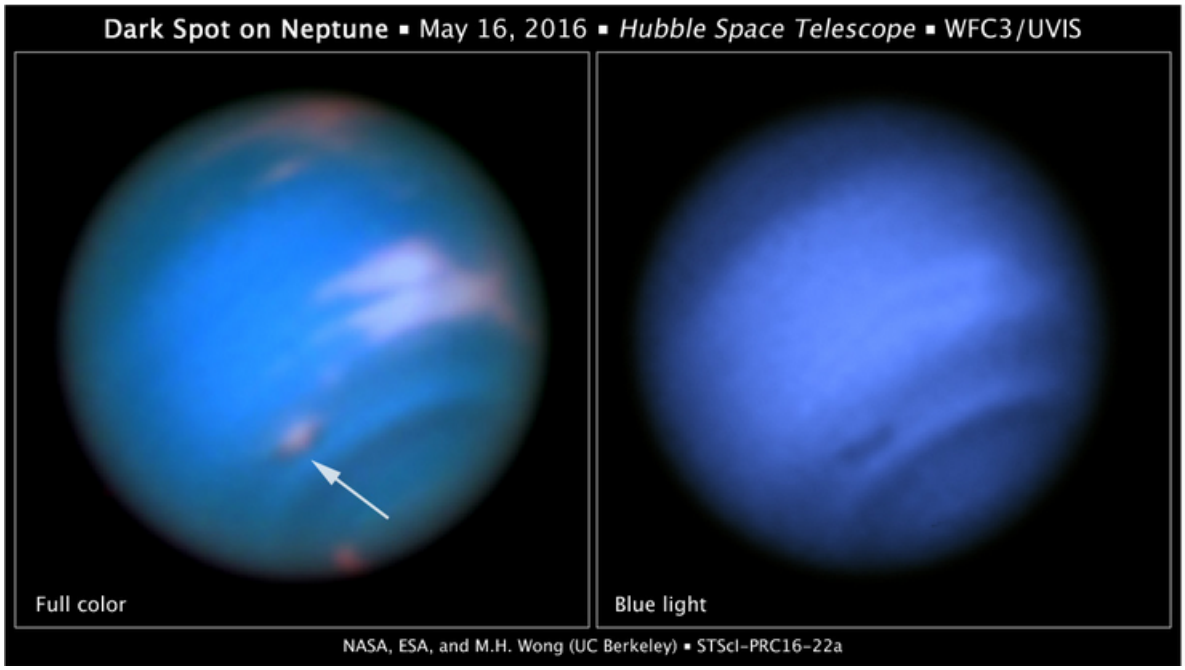

Figure 2: A dual image comparison of the planet Neptune from the Hubble Space Telescope, under Full colour (left) and Blue light (right), with the identified dark spot on the Full colour image by a white arrow. Taken from Aeronautics et al. (2016)

The original dark spot on Neptune has since disappeared; however, there have been

[1]The original mission also included Pluto, which was demoted to a dwarf planet in 2006 by the IAU (History.com, 2022).

findings of new dark spot-like vortex storms appearing in Neptune's atmosphere. The Hubble OPAL program identified a new dark spot with bright clouds that acted as companions to that storm in 2018 (Simon et al., 2019), which corresponds to the identified dark spot seen in Figure 2, and has been denoted the name "NDS-18", as seen in Simon et al. (2019). Whilst Simon et al. were unable to fully prove their hypothesis from the data only from OPAL, they were able to correlate the data that was determined by Voyager 2 and previous data from Jupiter missions, to find that the drift rate of the dark spot is approximately a match, the table of drift rate can be seen in table 1.

Table 1: Values of drift rate for the identified dark spot from each year, with the corresponding UTC and longitude. Taken from Simon et al. (2019).

Date	UTC	W. Longitude (deg)	drift rate (deg/hr)
2015-09-18	17.805	11	2.45
2016-10-04	0.558	87	2.46
2017-10-07	7.278	253	2.45
2018-09-10	2.160	339	2.44
2018-11-05	13.548	48	///
2018-11-06	8.622	95	2.46

As well as the drift rate, Simon et al applied some assumptions to the morphology of NDS-18 and found that the morphology suggested a vertical wind shear - how height changes the speed of wind (Hakim & Patoux, 2021). The assumptions applied by Simon et al involve the intrinsic vortex ζ_0 and the fact that NDS-18 follows the Kida equations for a stationary vortex (1) and (2) (Kida, 1981).

$$\zeta/\zeta_0 = (1-\lambda)/(\lambda(1+\lambda)) \tag{1}$$

$$\zeta = f(1-\lambda)/(2\lambda(1+\lambda)-(1-\lambda)) \tag{2}$$

The models of data provided by the OPAL program and Simon et al. (2019) show that there is a correlation between the vortex depth and height increase due to wind shear - suggesting that the formation and disappearance of the dark spots is due to the vortex drifting across the atmosphere.

2.3 Dark Spot - Current research and Understanding

This subsection will continue to discuss understanding and research into "NDS-18" (and other formed dark spots) and look at potential future missions that will study Neptune in

more detail.

Adding onto the models from Simon et al. (2019), Wong et al. (2022) found that by applying the Kida equations (Kida, 1981) to the vertical wind shear, producing Figure 3, which helped Wong et al. conclude that the wind shear of NDS-18 goes against the strong vertical wind shear that was implied by the Kida model in Simon et al. (2019). In doing this, along with other numerical simulations, Wong et al. concluded that the tangential velocity of NDS-18 is < 50 ms^{-1} which implied that the dark spot's intrinsic velocity is weak.

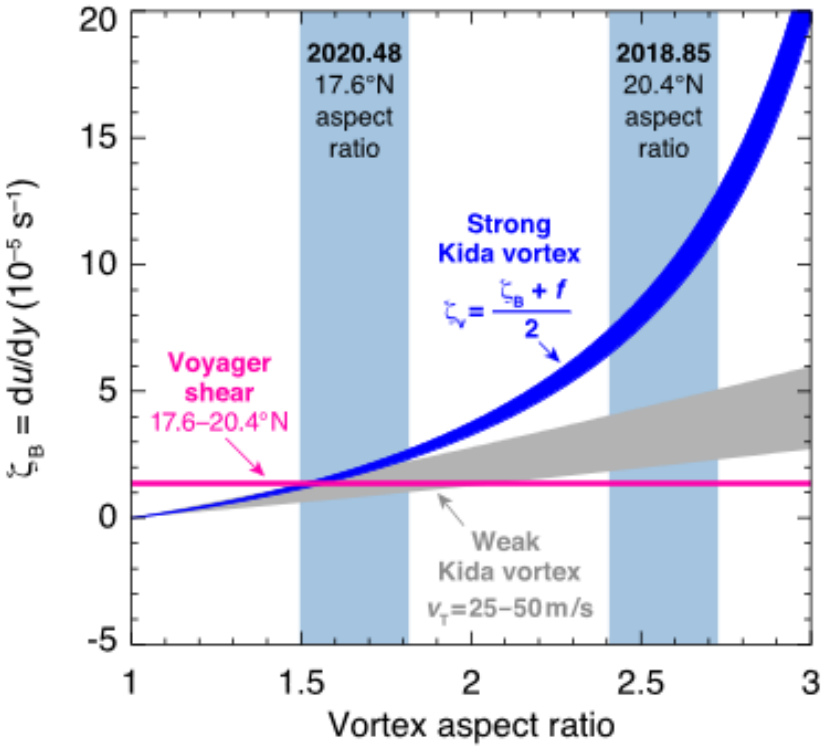

Figure 3: Observed vortex aspect ratio [light blue bands for two epochs] related to wind shear in the environment determined by Kida equations, producing the strong vortex curve [dark blue] or weak vortex curve [light grey] with a ν_T at a fixed 25-50 ms $^{-1}$ value.

As well as NDS-18, a deep bright spot labelled "DBS-2019" has been found that lies on the edge of NDS-18. Irwin et al. (2023) used spectral observations made by the Multi Unit Spectroscopic Explorer (MUSE) of the Chilean Very Large Telescope (VLT) in order to determine the vertical structure and apparent colour of the two spots. Irwin et al. determined that NDS-18 and DBS-2019 lie in close proximity to each other, which can be seen in Figure 4. The colours determined by this figure is denoted by Irwin et al. as a

"false-colour composite" with DBS-2019 showing orange in colour, and NDS-18 showing as a grey oval shape.

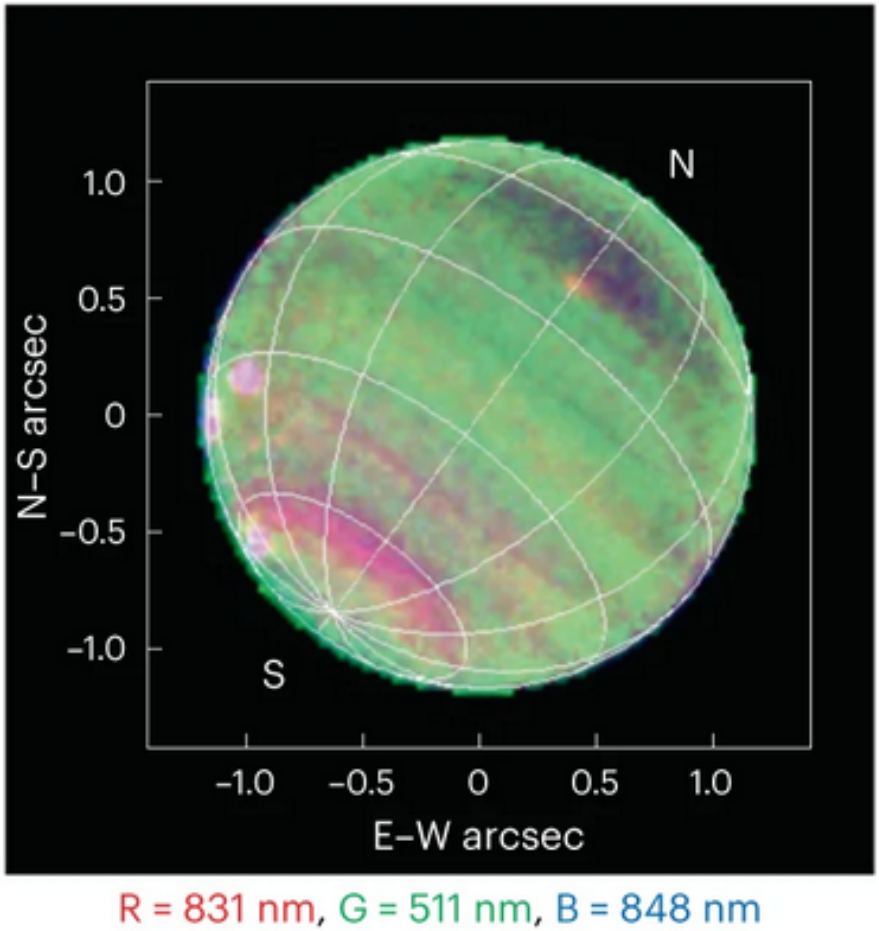

Figure 4: Image shows an apparent colour image of Neptune in three distinct colour band wavelengths, R = 831 nm, G = 511 nm and B = 848 nm. NDS-18 is displayed in a dark grey colour, and DBS-2019 can be seen in a bright orange colour. Taken from Irwin et al. (2023)

In 2020, Hadland et al. used the Explicit Planetary Isentropic Coordinate General Circulation Model (EPIC GCM) to investigate and simulate the evolution of vorticies on Neptune, like NDS-18. Hadland et al. (2020) used EPIC using an active cloud microphysics model, to produce a 100 day simulation of NDS-18. This produced Figures 5 and 6, which show a visual representation of a 60 day time frame of how the change of the longitude and latitude of the spot, and a graph of how the drift rate and inverted aspect ratio (semiminor axis $[b]$ over semimajor axis $[a]$: $\frac{b}{a}$) against a time span of 80 days.

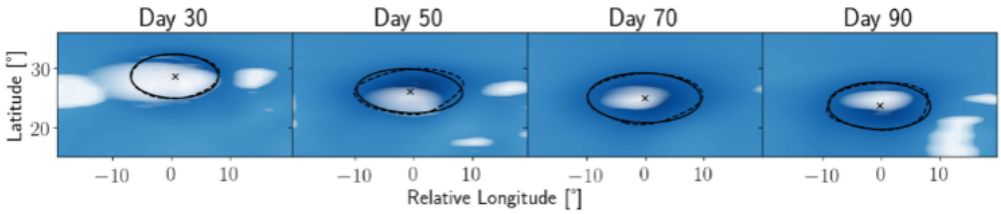

Figure 5: Four images of 20° longitude against a 10° latitude range for instantaneous times of the model on day 30, 50, 70 and 90. Images used the EPIC model with active microphysics for methane and a 40 × solar methane abundance, and plotted a dark blue dashed line of the vapour field contour, a solid line showing the fitted ellipse. Taken from Hadland et al. (2020).

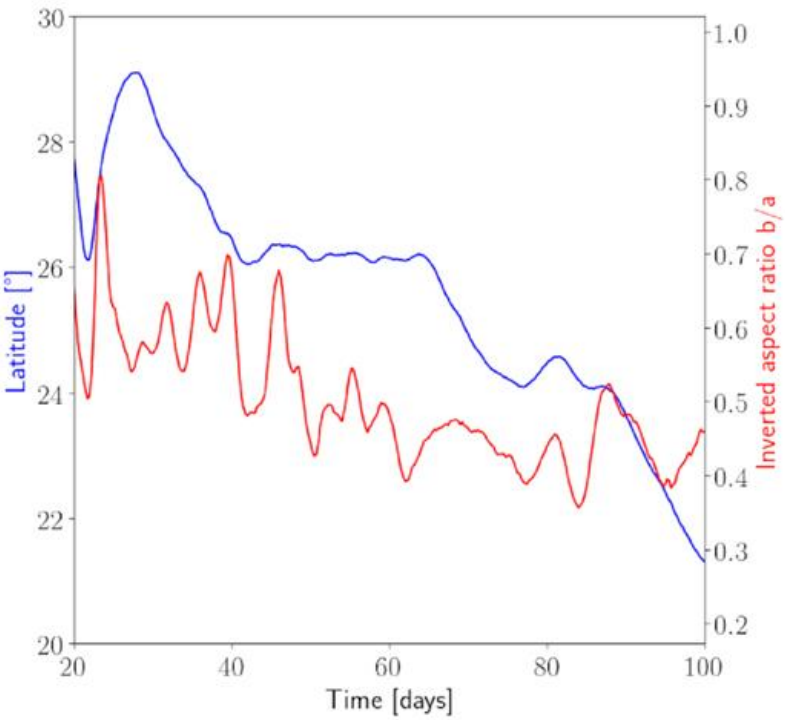

Figure 6: A graph of the drift rate [blue plot] and the aspect ratio [red plot] of the NDS-18 model as a function of the 100 day time frame. Taken from Hadland et al. (2020).

Using this model, Hadland et al. (2020) determined that the observed aspect ratio of NDS-18 was around $5.4 \times 10^{-5}\ s^{-1}$, that the dark spot became more elliptic over the 100 day period, and that there was more irregularity in the equatorial direction drift of the vortex model and had a higher drift rate.

McKevitt et al. (2024) produced an article outlining a new concept for a mission and

spacecraft that would observe the outer areas of the Solar System, like Neptune. The Arcanum mission is a proposed mission for operation near Neptune as it would allow for repeated orbits of the planet by accessing the areas with fastest winds and adverse conditions.

3 Mars: Dust Devils

3.1 Mars - Planetary Overview

Mars has a very thin and cold atmosphere compared to Earth, mainly comprised of CO_2 (around 95.3% by volume (Atreya & Gu, 1995)), little N_2 and ^{40}N (2.7% and 1.6% respectively), and has little water vapour (around an average of 150-200 ppm (Atreya & Gu, 1995)), due to the lower temperatures (Taylor, 2010).

The atmosphere of Mars appears generally transparent, with occasional dust storms that hide the planet's surface. Here, tiny dust particles of surface material are swept up if wind speeds exceed between 50 ms^{-1} and 100 ms^{-1}, which causes the material to strike the surface at high speeds and force even smaller particles into the atmosphere, where they can become suspended for weeks at a time. This process is known as "saltation". These dust storms are most common when the strongest surface winds are present, and Mars is near perihelion. (Moore, 2014). Some of these smaller storms that are prevalent on the Martian surface are "Dust Devils".

3.2 Definition and generation of dust devils

Dust devils are defined as a strong, violent, but small circulation, which is created when superheated and unstable dry air near a planet's surface breaks through the boundary layer and rises upward rapidly (NOAA & Cooley, 1971). For dust devils to form, three conditions need to be met. Firstly, a convective vortex must be produced (Rafkin et al., 2016), which is caused by the surface insolation of a planet leads to a super-adiabatic lapse rate (Balme & Greeley, 2006). Second, the produced vortex must have strength that is sufficient enough for the winds in the vortex to lift any sand or such particles into the air (Rafkin et al., 2016). The third condition is that there must be a source of surface material that is sufficient enough to produce a difference in visual opacity when the material is lifted (Rafkin et al., 2016).

When the vortex is created, an area of lower pressure at the core. As the dust devil moves, across the planet's surface rapidly, that pressure drops quickly - faster than the pressure can create an equilibrium with the pores that are inside the soil. That low pressure

then results in the grains in the surface to lift and rise into the vortex itself (Sweeney, 2022).

Many dust devils tend to form on flat surfaces of a planet, with weaker ambient winds (Mattsson et al., 1993), which means that another vorticity is required (Rafkin et al., 2016). These sources could come from the concentration of vorticity from the rotation of the planet, kilometre-scale swirls or the horizontal vorticity (the horizontal boundary layer vortices) into the vertical plane (Rafkin et al., 2016). Since dust devils appear to not show a preference for a particular direction of rotation, the rotation of the planet is unlikely to be the source of the dust devil vorticity. This is shown by the Rossby number, R_o, which is the ratio of Coriolis force (the apparent force experienced by an object that is moving in a rotating system) and inertial force for a system (Rafkin et al., 2016). The equation for Rossby number can be seen in equation 3.

$$R_o = \frac{V}{2\Omega L} \tag{3}$$

Where V is flow speed (SI units ms^{-1}), Ω is the planet's rotational angular velocity in the vertical direction, and L is the length characteristic of the flow. In dust devils that are very large, the R_o value is >1, which implies that the Coriolis forces are insignificant (Rafkin et al., 2016).

3.3 Current Understanding of Mars' dust devils

Multiple land based explorations have been conducted on Mars, like Pathfinder in 1996 and Phoenix in 2007. Lorenz (2012) used the data from these two missions, to produce a series of figures that implement a power law distribution to the pressure drops of dust devils found on mars, and compared those distributions to those of terrestrial dust devils, seen in Figure 7. The terrestrial pressure drop dust devil data came from Lamberth (1966) and their research into dust devils around the White Sands Missile Range (WSMR) in the southern New Mexico desert, the data from Pathfinder from Murphy & Nelli (2002) and the data from Phoenix from Ellehoj et al. (2010).

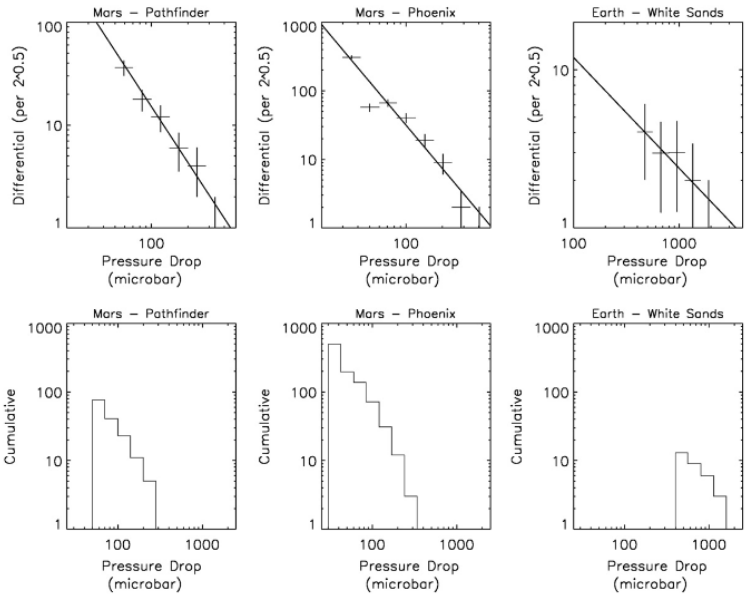

Figure 7: From left to right, pannels show Mars Pathfinder data (Murphy & Nelli, 2002), Mars Phoenix data (Ellehoj et al., 2010) and White Sands data (Lamberth, 1966). The top panels show the events counted in a $2^{0.5}$ logarithmic bin with horizontal bars of width with counting errors of $N^{0.5}$ bars in the vertical direction, against pressure drop (μbar). Panels also show a power law fit of -1.56 $\pm$ 0.49, -2.20 $\pm$ 0.47 and -0.76 $\pm$ 0.74. Lower panels show cumulative counts against pressure drop (μbar). Taken from Lorenz (2012)

Lorenz (2012) found the Pathfinder and Phoenix data were consistent with an exponent power law of -2 shown in Lorenz's earlier work on dust devil diameter in Lorenz (2009), showing that both Mars and terrestrial data could be consistent with a -2 power law - but further work would be needed to definitively prove this claim.

In 2016, Murphy et al. (2016) used the proposed power -2 law to compare the pressure drop created by dust devils during the formation process ,ΔP, of Mars Pheonix data, and the data of multiple teams around El Dorado in Nevada, USA. This comparison is seen in Figure 8.

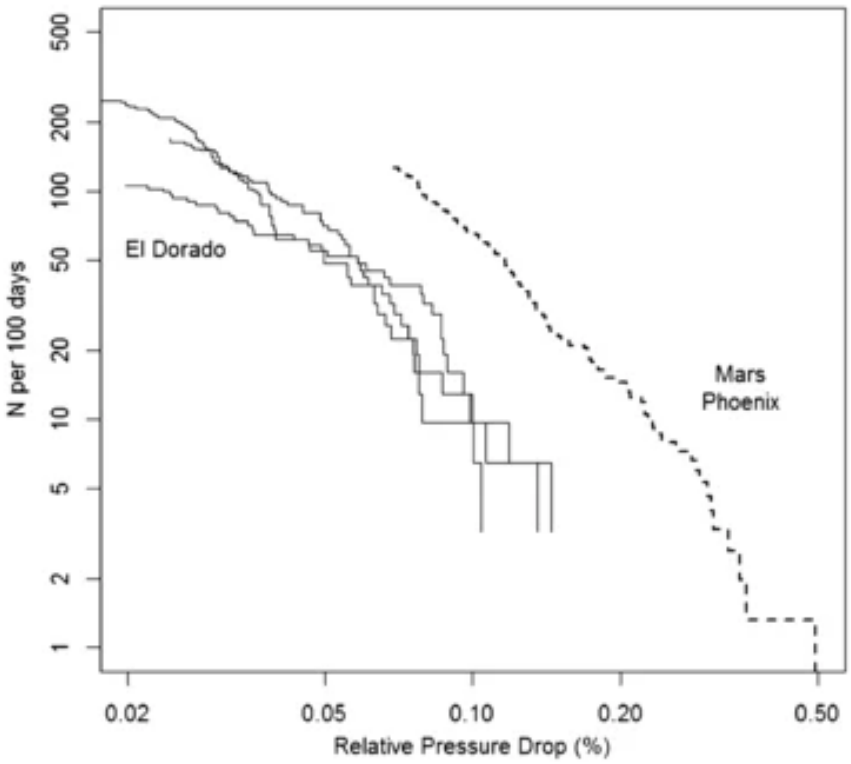

Figure 8: Percentage relative Pressure drop comparison of Mars Phoenix (dashed line plot) and El Dorado (straight line plot) dust devil data, using a power -2 law distribution. Taken from Murphy et al. (2016)

Murphy et al. (2016) determined using Figure 8, that terrestrial dust devils correspond to approximately a 0.1 % (or 1 mbar) pressure drop, whilst dust devils on Mars is approximately 0.3 % (around 0.02 mbar). This decrease in pressure, while surrounding a fraction of a percent, the lowered pressure drop of Martian dust devils could be due to absolute pressures of Mars being that of over 100 times smaller than on Earth.

Using a basic understanding of ΔP, Balme & Greeley (2006) determined a derivation from the pressure drop in a convective vortex to obtain the peak tangential wind speed V_{max}, the derivation can be seen through equation 4.

$$V_{max} \approx (\frac{\gamma \eta c_P \Delta T}{P_\infty})^{1/2} \tag{4}$$

Where η is the fraction of work converted from heat input, γ is the fraction of mechanical energy consumed by friction near the ground, c_P is the specific heat capacity at a constant pressure of the atmosphere, P_∞ is the pressure away from circulation influence, and ΔT is the temperature perturbation for convective plumes over homogenous surfaces. With this

determination of a max tangential velocity, and the knowledge that the Martian surface has a thinner surface pressure (approximately 5.2 mbar to Earth's 1000 mbar (Gray Young, 1971)) Balme & Greeley (2006) produced Figure 9, which compares the threshold friction wind speed required by Martian winds to lift smaller particles to form a dust devil.

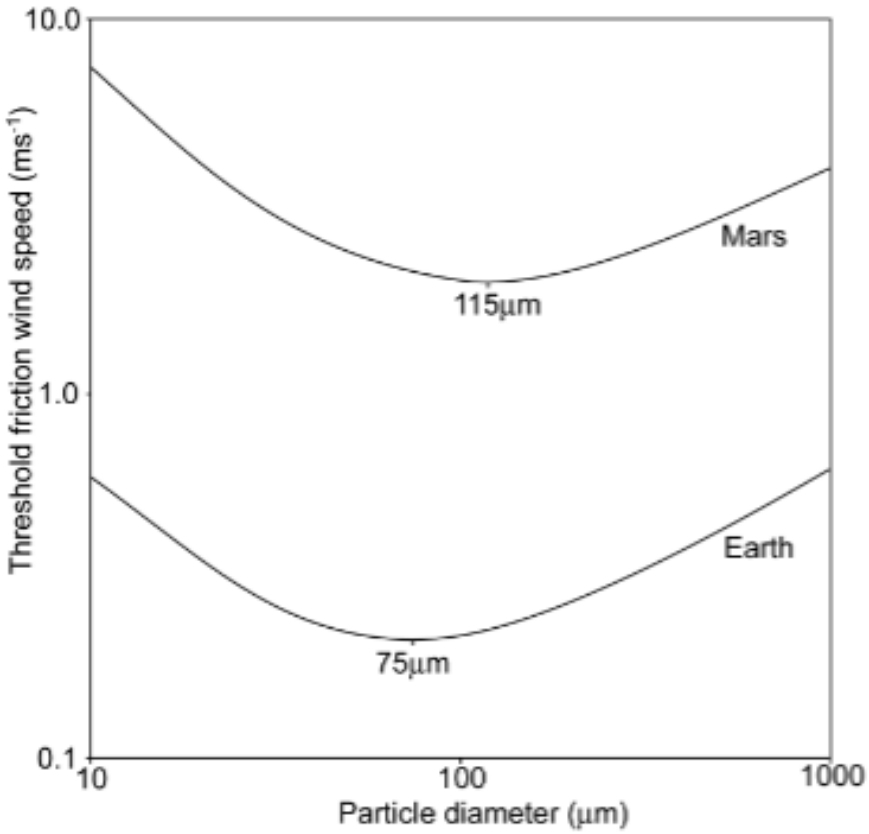

Figure 9: Threshold friction wind speed (ms^{-1} for Mars and Earth materials of 115 μm and 75 μm diameters. Taken from Balme & Greeley (2006).

This constant pick up of material on Mars has led to images being produced of track-like paths on the Martian surface. The amount of material D removed from dust devils can be determined using equation 5.

$$D \approx \frac{1}{\alpha}[N_t \bar{A}_t \bar{m}_t] \tag{5}$$

Where α is the fraction of all dust devils that leave tracks. N_t is the number of tracks formed on Mars per year, $\bar{A}_t$ is the average dust devil track area and $\bar{m}_t$ is the mean mass of material removed per unit of area to form the track. Taken from Balme & Greeley (2006). Balme & Greeley (2006) suggested that the dust devils on Mars could be the

dominant process in maintaining the opacity of the atmosphere.

4 Jupiter: Thunderstorms and Lightning

4.1 Jupiter - Planetary Overview

Jupiter is the largest planet in our solar system, with a mass of 317.89 $M_\oplus$ and an equatorial radius of 71,942 km (Moore, 2014). While Jupiter has been visible using known about for centuries, the scientist denoted for the first detailed observations is Galileo Galilei in 1610 (Pasachoff, 2015). Various missions have given data on the atmospheric composition of the Jovian planet (like Juno which will be discussed in a later subsection) which have all shown that H_2 dominates Jupiter's atmosphere, followed by He and a mix of H_2O, CH_4, H_2S and NH_3 (Atreya et al., 2003) making up the different vertical atmospherical structure. The inside of the troposphere contains a complex cloud system of water, ammonia and ammonium hydrosulphide (Atreya & Wong, 2005). Aglyamov et al. (2021) suggested that underneath these ammonia clouds there is a thin layer of water clouds, as was found by Atreya & Wong (2005), and that within these clouds of water there have been small flashes of lightning. This idea of lightning generation is discussed in the next section.

4.2 Generation of thunderstorms and lightning

Lightning generation, as we currently understand it, is generated in a way similar to that on Earth. This section will focus on how lightning and thunderstorms are known to generate on Earth. To look at lightning in detail, an understanding of how cumulonimbi are formed must be understood.

4.2.1 Cumulonimbus cloud formation

At the beginning of formation, cumulus clouds in the atmosphere mix with dry air. This process is called entrainment, which culminates in the evaporation of water droplets into the cloud mixture, due to the reduction in the dew point temperature (Hakim & Patoux, 2021). This causes moisture to be transported upward from the surface to higher areas of increased humidity. Even if some clouds dissipate, there is a net increase of moisture from entrainment, which reduces the evaporation of the next cloud that goes through the entrainment process. After some time, the cumulus cloud may be tall enough that the cloud droplets within them make precipitation which is often large drops of rain. This is a relatively fast process that creates conductive raindrops, and if the cumulus cloud

reaches heights where -10°C temperatures are hit, then ice crystals start to form (Hakim & Patoux, 2021). This formation of ice crystals will become more relevant in the next section. When the cumulus cloud reaches the stratosphere or a layer of the troposphere that is stable, it flattens against that layer and is now a cumulonimbus. An image of this anvil-like appearance can be seen in Figure 10 (Hakim & Patoux, 2021).

Figure 10: A photograph of a cumulonimbus cloud in the anvil shape once it reaches the stratosphere or stable troposphere, taken from Hakim & Patoux (2021).

The resulting cumulonimbus cloud forms an ordinary thunderstorm.

Any precipitation falling into sub-saturated air below the cloud base evaporates, and since evaporation requires latent heat taken from the environment this has a cooling effect. As cold air is more dense, it begins to sink down under the cloud which creates cool downdrafts that compete against more convective updrafts. The friction between the falling precipitation and the air surrounding them aids in enhancing the downdrafts as it drags the air downwards (Hakim & Patoux, 2021). When the cloud reaches the surface, a cold pool is formed which spreads out and creates a gust of wind. As the cold pool cuts off the warm humid air to the storm, the storm dissipates. At the same time, the cold air goes against the warm air, and the boundary created is called a gust front. Like the cold front, the gust front forces an upward motion which may create small cumulus clouds - which can start the process of creating a cumulonimbus cloud again and develop a new ordinary thunderstorm (Hakim & Patoux, 2021).

4.2.2 Lightning generation

As discussed in the previous subsection, small and large ice crystals form in the creation of a thunderstorm. If the small crystals ascend in the updrafts, then the larger falling crystals become negatively charged and act as an anode (and consequently, the smaller crystals become positively charged and act as a cathode) (Hakim & Patoux, 2021). To initiate

an electric discharge (at sea level) an approximate value of critical electric field of 3×10^6 V/m. The critical electric field is lowered if there is the presence of ice crystals in the thundercloud by a factor of around 500 kV/m. If this value is reached and exceeded, then electron avalanche begins and this process starts electrical breakdown (Cooray, 2015).

When an electrical breakdown begins, it gives rise to streamer charges, which then results in a leader discharge. When a leader is created, it propagates in the electric field, leading to a cloud or ground flash. This electrical breakdown forces a negative charge to travel through the cloud in successive pokes which extend in steps at each poke, this is known as a stepped leader, which moves in a rapid microsecond order and each of the steps gives a small flash. The opposing positive charge moves up from a positively charged area to meet the steps and build a path for more electrons to flow. This flow of electrons resulting from the connection is called the return stroke, which heats up the air to approximately 30 kK which appears to the eye as a white light. Rapid heating of the air surrounding the strike creates a shock wave that travels at the speed of sound - this is known as thunder (Hakim & Patoux, 2021).

When the channel of electron flow is established, and if the cumulonimbus cloud still contains pockets of negative charges, more electricity can flow through the return stroke. Any new flow of electrons on the established channel is the dart leader, which causes more return strokes (Hakim & Patoux, 2021). These newly created return strokes all use the same main-channel lightning strike from the cloud to the positively charged area of the strike.

4.3 Lightning on Jupiter - Current research and Understanding

This section aims to discuss current research on Jupiter's lightning and the understanding of lightning from past space agency missions.

Lightning has been found on the Jovian planet by multiple space agency missions, first thought to be found by Voyager 1 detecting radio signals (known as whistlers) in 1979 (Gurnett et al., 1979). Gibbard et al. (1995) used data from the Voyager 1 mission to calculate and create water cloud models for varying pressure levels; 2, 3, 4 and 5 bar, and estimated the break down of electric field with respect to time to compare the development of lightning on Jupiter to that of terrestrial lightning. This model can be seen in Figure 11.

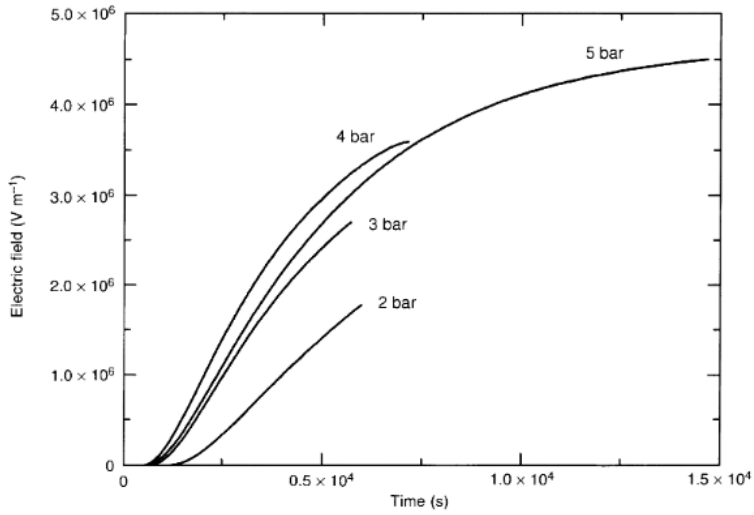

Figure 11: The development of a large scale electric field in a jovian thunderstorm, the electric field [V m^{-1}] as a function of time [s] for 2,3,4 and 5 bar pressures. Taken from Gibbard et al. (1995).

To aid in the comparison, Gibbard et al. also used the model to show how the electric field strength varied with water abundance, all at 5 bar pressure, for water abundances at the standard, half and double values, once again, with respect to time. This can be seen in Figure 12.

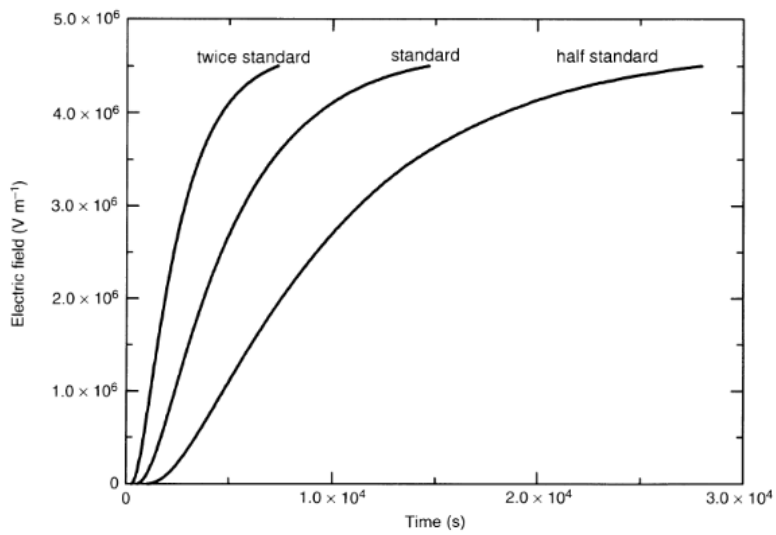

Figure 12: Temporal development of electric field [V m^{-1}] at a 5 bar pressure for water abundances at standard, half and double the baseline values. Taken from Gibbard et al. (1995).

From these two figures, Gibbard et al. determined that Jovian lightning is more likely to occur at 3 or 4 bar pressure levels, where the temperature allows for a short enough timescale that the breakdown field levels can be achieved before water particles are able to be rained out through precipitation. In addition to this, they go on to say that as the 5 bar level is approached, the electrostatic lifting of the precipitating particles slows down the electrification so much that the water particles rain out before the electron breakdown is achieved. However, this is not true unless the water abundance is higher than the baseline.

Gibbard et al. (1995) used these models to aid further observation by the then upcoming Galileo mission. While the Voyager 1 mission managed to detect radio signals only *thought* to be lightning on Jupiter, Galileo's SSI (Solid State Imager) module managed to get photographic evidence that lightning *does* occur on the gas giant. Little et al. (1999) showed various of these images, like Figures 13 and 14 that show lightning on Jupiter's night side roughly 75 minutes apart, and three storms at different times.

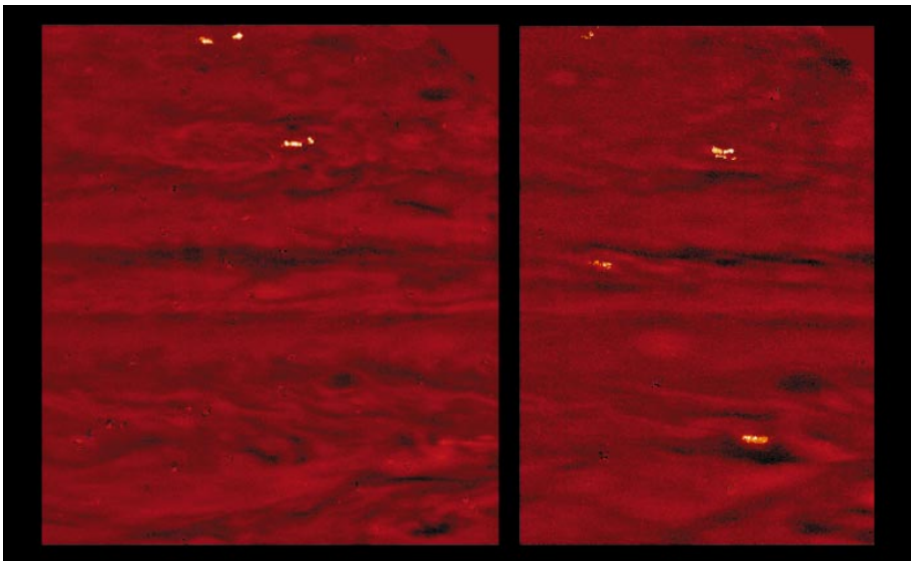

Figure 13: Images show white lightning storms against red and black backgrounds of clouds lit up by Io, taken 75 minutes apart. [LEFT: storms 7, 8 and 10] [RIGHT: storms 5, 11, 13 and 15] Taken from Little et al. (1999).

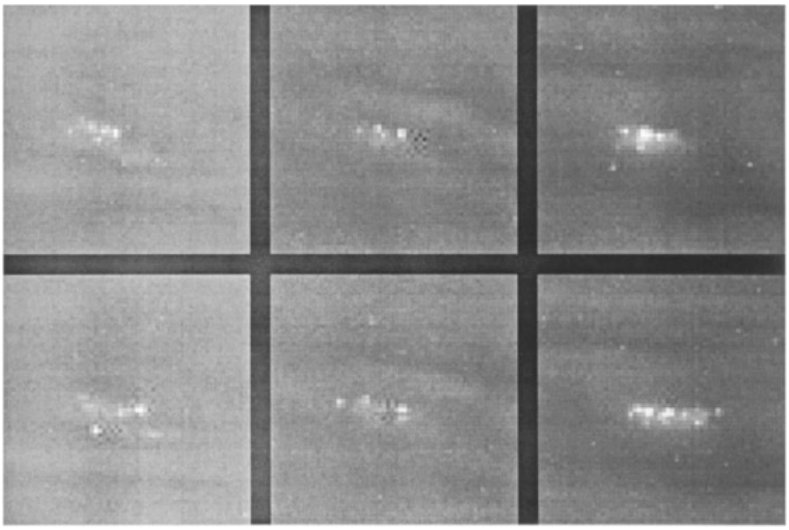

Figure 14: Image shows three storms (11, 13 and 15 respectively) at two different times approximately 2 minutes apart taken in a 89.6s exposure. Taken from Little et al. (1999).

The most recent (and ongoing) mission to Jupiter is the Juno spacecraft (NASA et al., 2025), launched in 2011. Using data from Juno's SRU (Stellar Reference Unit), Becker et al. (2020) confirmed that shallow lightning flashes can be seen in multiple atmospheric regions, and a comparison of flash number against optical energy [J] of Juno data along with previous mission data can be seen in Figure 15.

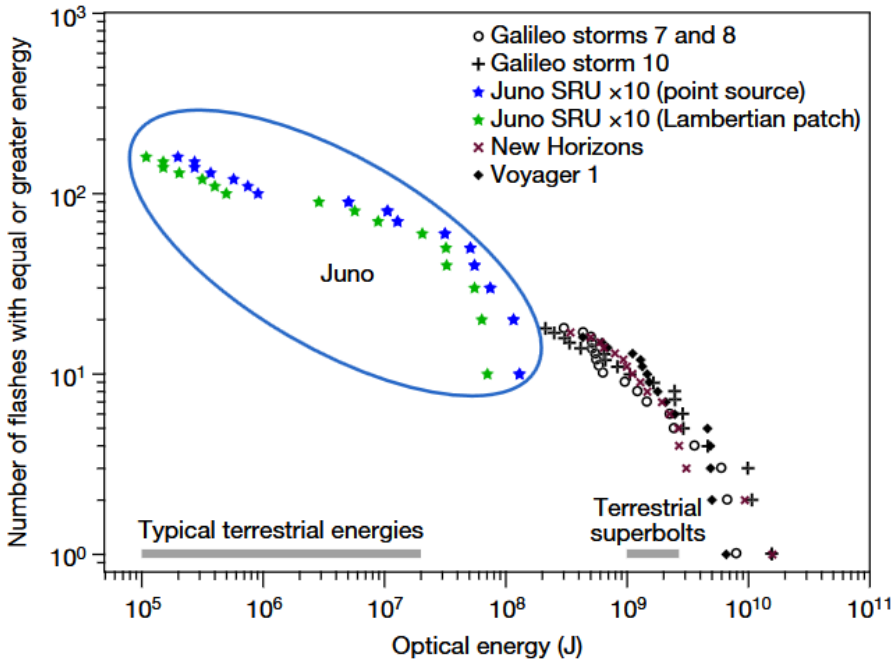

Figure 15: Flash energy data from various instruments from past and current missions plotted as a frequency distribution against optical energy [J], taken from Becker et al. (2020).

This was further expanded on by Aglyamov et al. (2021) who identified that the generation of lightning on Jupiter requires the ammonia in it's atmosphere to stabilise liquid water at sub-freezing altitudes. They also found a range of water abundances at which lightning generation is possible using Juno data - which reflects the models put forward by Gibbard et al. (1995), over 20 years earlier.

Looking further at Juno, Kolmašová et al. (2023) found that lightning on Jupiter strikes a rhythm similar to that of cloud lightning on Earth. Looking particularly at return stroke speed, Kolmašová et al. (2023) found that the return stroke cadence on Jupiter is around 1 μs. Similarly, Chen (2018) determined that in assumed constant return stroke speed of $\frac{1}{3}$ of the speed of light, at an average strike distance of 100m, a return stroke cadence of around 1.1 μs.

5 Conclusion

In conclusion, there is a clear link between the formation of different weather phenomena on the other planets in the solar system, and how they are formed on Earth. Understandably, there is more known and researched into the planets closer to Earth like Mars and Jupiter, than there is on the outer planets like Neptune. However, this just poses the potential for more in depth missions to be completed into planets past Jupiter and Saturn, like the previously mentioned Arcanum mission from McKevitt et al. (2024) to Neptune and it's moons.

The questions "What weather phenomena happen within the Solar System?" and "What research has been done into these phenomena?" which were set out in the introduction of this dissertation were believed to have been answered, showing an in depth analysis of the generation of the Great Dark Spot on Neptune, dust devils on Mars and lightning on Jupiter, as well as how current and previous research has developed the understanding of the phenomena.

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